



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2009/2010

INSTITUTION BASED PROGRAMME

APRIL SESSION EXAMINATION FOR THE DEGREE OF MASTER OF EDUCATION

ECT 534: APPLICATION OF MATHEMATICS IN DAILY LIFE

DATE: Friday 3rd September, 2010

TIME: 9.00 a.m. – 12.00 p.m.

INSTRUCTIONS

Answer THREE Questions including question 1.

1.
 - (a) What is the nature of mathematics and how do teacher expectations of learners differ by this dual nature of mathematics? [10 marks]
 - (b) What are the goals for mathematics education with respect to knowledge, skills and attitude? [10 marks]
 - (c) Besides being an essential element of communication, mathematics is known for its utility. Discuss this value of mathematics in our daily lives. [10 marks]

2.
 - (a) Behaviorism and constructivism theories of learning apply to mathematics education. Describe what these theories are and highlight their strengths and weaknesses for teaching and learning mathematics in Kenyan school system.
 - (b) What are some of the potential roles of a computer in the classroom? For each role, identify value and limitation. [15 marks]

3.
 - (a) Mathematics has been described as neither mental nor physical, it is abstract. Give some examples to support this assertion.

- (b) What is the role of mathematics in National development? Consider at least four different sectors of the national economy. [15 marks]
4. When students are asked about their views about mathematics, they state that “it is difficult” further probing always results in identification of *inappropriate learning environment*. Discuss this view. How would you relate teachers’ attitude towards mathematics to student learning? [15 marks]
5. (a) In your opinion, why do people make mistakes in mathematics?
- (b) What does it mean to understand conceptually and how can we help pupils modify their understanding. Give examples from Kenyan school mathematics curriculum. [15 marks]

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